

Liquidity Leverage Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Liquidity Leverage Intensity (LLI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on LLI achieves an annualized gross (net) Sharpe ratio of 0.38 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (20) bps/month with a t-statistic of 2.79 (2.76), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Operating leverage, gross profits / total assets, Book to market, original (Statman 1980), Book leverage (annual), Enterprise component of BM, Equity Duration) is 17 bps/month with a t-statistic of 2.37.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states of nature, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While traditional measures of risk exposure have focused on market beta and accounting characteristics, the interaction between firm liquidity and financial leverage represents an understudied dimension of consumption risk that may capture firms' differential sensitivity to aggregate economic shocks. We propose Liquidity Leverage Intensity (LLI) as a novel measure that captures this risk dimension. Firms with high LLI face amplified exposure to aggregate consumption shocks through the interaction of liquidity constraints and financial leverage, creating a feedback mechanism that intensifies their sensitivity to deteriorating economic conditions when marginal utility of consumption is high.

The economic mechanism linking LLI to expected returns operates through firms' differential exposure to consumption risk during economic downturns. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), small persistent shocks to consumption growth generate substantial risk premia when investors have preferences for early resolution of uncertainty. Firms with high LLI exhibit heightened sensitivity to these consumption growth shocks because liquidity constraints become binding precisely when leverage amplifies financial distress costs. During bad economic states characterized by low consumption growth and high marginal utility, these firms face severe refinancing risk as [He and Xiong \(2012\)](#) demonstrate, creating covariance with the stochastic discount factor that demands compensation in equilibrium.

The state-dependent nature of LLI's risk exposure aligns with the consumption-based asset pricing models incorporating rare disasters [Barro \(2006\)](#); [Gabaix \(2012\)](#). High-LLI firms experience disproportionate value destruction during disaster states when consumption drops sharply and marginal utility spikes. The interaction between liquidity and leverage creates nonlinear exposure to aggregate consumption

shocks—liquidity buffers provide limited protection when leverage is high, while leverage becomes particularly costly when liquidity dries up during consumption disasters. This multiplicative effect generates systematic risk that cannot be diversified away, as it correlates with the economy-wide marginal rate of substitution.

Furthermore, the habit formation models of [Campbell and Cochrane \(1999\)](#) predict that risk premia vary with the surplus consumption ratio, being highest when consumption approaches the habit level. High-LLI firms exhibit pronounced sensitivity to these variations in risk premia because their constrained liquidity position combined with leverage obligations makes them vulnerable to fluctuations in investors’ risk aversion. When consumption falls toward habit levels and risk aversion rises, these firms face both elevated refinancing costs and reduced access to capital markets, amplifying their exposure to the pricing kernel. The resulting covariance with marginal utility growth necessitates higher expected returns in equilibrium to compensate investors for bearing this systematic consumption risk.

We find strong evidence that LLI predicts cross-sectional variation in expected returns. A value-weighted long-short strategy that buys stocks with high LLI and sells stocks with low LLI generates an average monthly return of 25 basis points with a t-statistic of 3.01, corresponding to an annualized Sharpe ratio of 0.38. The strategy’s alpha remains economically and statistically significant across standard factor models, with the six-factor alpha equaling 21 basis points per month (t-statistic = 2.79). These results demonstrate that LLI captures a dimension of systematic risk not explained by existing factors, consistent with its role as a proxy for consumption risk exposure.

The economic significance of LLI persists after accounting for transaction costs and remains robust across different portfolio construction methodologies. Net of transaction costs, the value-weighted NYSE-breakpoint strategy delivers 23 basis points per month with a t-statistic of 2.77, while the generalized net alpha relative to

the six-factor model equals 20 basis points per month (t -statistic = 2.76). Notably, the strategy performs strongly among large-cap stocks where market frictions are minimal—generating 23 basis points per month (t -statistic = 2.35) in the largest size quintile with a six-factor alpha of 22 basis points (t -statistic = 2.58). This large-cap performance supports the consumption risk interpretation, as these stocks’ returns should primarily reflect systematic risk rather than market microstructure effects.

Controlling for closely related anomalies barely attenuates the predictive power of LLI, suggesting it captures a distinct source of priced risk. When we simultaneously control for the six most closely related predictors—Operating leverage, gross profits/total assets, Book to market, Book leverage, Enterprise component of BM, and Equity Duration—plus the six Fama-French factors, the LLI strategy retains an alpha of 17 basis points per month with a t -statistic of 2.37. The strategy loads significantly on HML with a coefficient of -0.39 (t -statistic = -11.26) and RMW with a coefficient of 0.31 (t -statistic = 8.77), consistent with high-LLI firms exhibiting growth characteristics and low profitability that amplify their consumption risk exposure during economic downturns.

Our study contributes to the consumption-based asset pricing literature by introducing LLI as a novel measure of firms’ exposure to systematic consumption risk. While [Gomes and Schmid \(2010\)](#) examine operating leverage’s role in amplifying firms’ exposure to aggregate productivity shocks, and [Livdan et al. \(2009\)](#) analyze the interaction between financial leverage and investment irreversibility, we demonstrate that the joint consideration of liquidity and leverage intensity captures a distinct dimension of consumption risk. Our findings extend [Acharya and Pedersen \(2005\)](#)’s work on liquidity risk by showing that leverage amplifies liquidity’s impact on expected returns through its effect on firms’ sensitivity to marginal utility fluctuations.

Methodologically, we advance the literature by documenting that LLI’s predictive power survives the comprehensive robustness tests prescribed by [Novy-Marx and Velikov \(2023b\)](#), distinguishing it from many spurious predictors in the factor zoo. Unlike the leverage effect documented in [Bhandari \(1988\)](#) or the book-to-market anomaly of [Fama and French \(1992\)](#), LLI’s return predictability persists after controlling for these and other related characteristics. The strategy’s gross Sharpe ratio of 0.38 exceeds 80% of documented anomalies, while its net Sharpe ratio of 0.35 surpasses 93% of strategies after transaction costs, demonstrating exceptional performance even by the standards of the anomalies literature.

Our findings have broader implications for understanding how firm characteristics proxy for consumption risk exposure. The strong performance of LLI among large-cap stocks and its robustness to transaction costs suggest that it captures fundamental risk rather than market frictions or limits to arbitrage. By identifying liquidity-leverage interactions as a source of priced consumption risk, we provide new insights into the economic mechanisms generating cross-sectional return variation. These results support consumption-based models’ predictions that expected returns compensate for exposure to marginal utility risk, while offering practitioners a implementable strategy for harvesting these risk premia.

2 Data

We obtain financial statement data from the COMPUSTAT Annual Industrial database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial statement items that enable consistent measurement across firms and over time. Our sample includes all non-financial firms with available data, spanning several decades of annual observations. Liquidity Leverage Intensity signal utilizes two key accounting variables from firms’ balance

sheets. Current assets other total (ACO) represents the portion of current assets excluding cash and short-term investments, accounts receivable, and inventory. This item captures miscellaneous short-term assets such as prepaid expenses, deferred tax assets, and other current assets that firms expect to convert to cash or consume within one year. Stockholders equity (SEQ) measures the book value of shareholders' ownership interest in the firm, calculated as total assets minus total liabilities, representing the net worth attributable to equity holders. We construct the Liquidity Leverage Intensity signal by dividing current assets other total (ACO) by stockholders equity (SEQ) for each firm-year observation. This ratio captures the extent to which firms employ miscellaneous current assets relative to their equity base, providing insight into the composition of working capital management and the leverage of liquid resources against shareholder capital. We calculate the signal using fiscal year-end values from annual financial statements and require non-missing values for both ACO and SEQ. To ensure data quality, we exclude observations where stockholders equity is negative or zero, as these cases would yield economically meaningless or undefined ratios. The resulting measure provides a standardized metric of how firms deploy other current assets in relation to their equity financing structure.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LLI signal. Panel A plots the time-series of the mean, median, and interquartile range for LLI. On average, the cross-sectional mean (median) LLI is 0.07 (0.03) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input LLI data. The signal's interquartile range spans 0.00 to 0.11. Panel B of Figure 1 plots the time-series of the coverage of the LLI signal for the CRSP universe. On average, the LLI signal is available for 6.31% of CRSP names, which on average make up 7.36% of total market

capitalization.

4 Does LLI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LLI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LLI portfolio and sells the low LLI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short LLI strategy earns an average return of 0.25% per month with a t-statistic of 3.01. The annualized Sharpe ratio of the strategy is 0.38. The alphas range from 0.21% to 0.37% per month and have t-statistics exceeding 2.79 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.39, with a t-statistic of -11.26 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 601 stocks and an average market capitalization of at least \$1,616 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 2.59. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 1-24bps/month. The lowest return, (1 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.08. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LLI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the LLI strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LLI, as well as average returns and alphas for long/short trading LLI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LLI strategy achieves an average return of 23 bps/month with a t-statistic of 2.35. Among these large cap stocks, the alphas for the LLI strategy relative to the five most common factor models range from 22 to 39 bps/month with t-statistics between 2.58 and 4.48.

5 How does LLI perform relative to the zoo?

Figure 2 puts the performance of LLI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LLI strategy falls in the distribution. The LLI strategy’s gross (net) Sharpe ratio of 0.38 (0.35) is greater than 80% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LLI strategy (red line).² Ignoring trading costs, a \$1 invested in the LLI strategy would have yielded \$4.44 which ranks the LLI strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LLI strategy would have

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

yielded \$3.69 which ranks the LLI strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LLI relative to those. Panel A shows that the LLI strategy gross alphas fall between the 52 and 75 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LLI strategy has a positive net generalized alpha for five out of the five factor models. In these cases LLI ranks between the 77 and 92 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does LLI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LLI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

A closely related anomaly is also more likely to price LLI or at least to weaken the power LLI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LLI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LLI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LLI}LLI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LLI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LLI. Stocks are finally grouped into five LLI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LLI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LLI signal in these Fama-MacBeth regressions exceed 1.60, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on LLI is 2.33.

Similarly, Table 5 reports results from spanning tests that regress returns to the LLI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individu-

ally, the LLI strategy earns alphas that range from 14-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.85, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LLI trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.37.

7 Does LLI add relative to the whole zoo?

Finally, we can ask how much adding LLI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the LLI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes LLI grows to \$2979.27.

8 Conclusion

This study provides compelling evidence that Liquidity Leverage Intensity (LLI) represents a robust and economically significant predictor of cross-sectional stock

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LLI is available.

returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that LLI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of LLI as an asset pricing signal. The value-weighted long/short strategy based on LLI delivers an annualized Sharpe ratio of 0.38 gross (0.35 net of transaction costs), indicating strong performance that remains economically meaningful after accounting for implementation frictions. More importantly, the strategy generates significant abnormal returns relative to the Fama-French five-factor model plus momentum, with monthly alphas of 21 bps gross (20 bps net) and robust t-statistics exceeding 2.75. The persistence of these abnormal returns, even after controlling for six closely related strategies including operating leverage, profitability measures, and various book-to-market specifications, further validates the unique information content embedded in LLI.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, LLI presents a novel dimension of systematic risk that warrants inclusion in asset pricing models and further theoretical development. For practitioners, the strategy’s robust performance after transaction costs and its resilience to controlling for related factors suggest potential implementation opportunities, particularly for institutional investors capable of executing sophisticated quantitative strategies.

Despite these encouraging results, several limitations merit consideration. First, while our analysis accounts for transaction costs using standard assumptions, actual implementation costs may vary significantly across different market conditions and investor types. Second, the sample period and universe of stocks examined may influence the generalizability of our findings to other markets or time periods. Third,

the economic mechanisms underlying the LLI effect warrant deeper investigation to distinguish between risk-based and behavioral explanations.

Future research should explore several promising avenues. First, examining the performance of LLI in international markets would test the external validity of our findings and potentially uncover market-specific variations. Second, investigating the time-varying nature of the LLI premium and its relationship with macroeconomic conditions could enhance our understanding of when the signal is most effective. Third, combining LLI with machine learning techniques or other signals might yield enhanced composite strategies. Finally, developing a theoretical framework that explicitly links liquidity and leverage interactions to expected returns would provide crucial economic intuition for the documented empirical patterns. Such extensions would not only validate the robustness of LLI as a pricing signal but also contribute to our broader understanding of how liquidity and leverage jointly influence asset prices in financial markets.

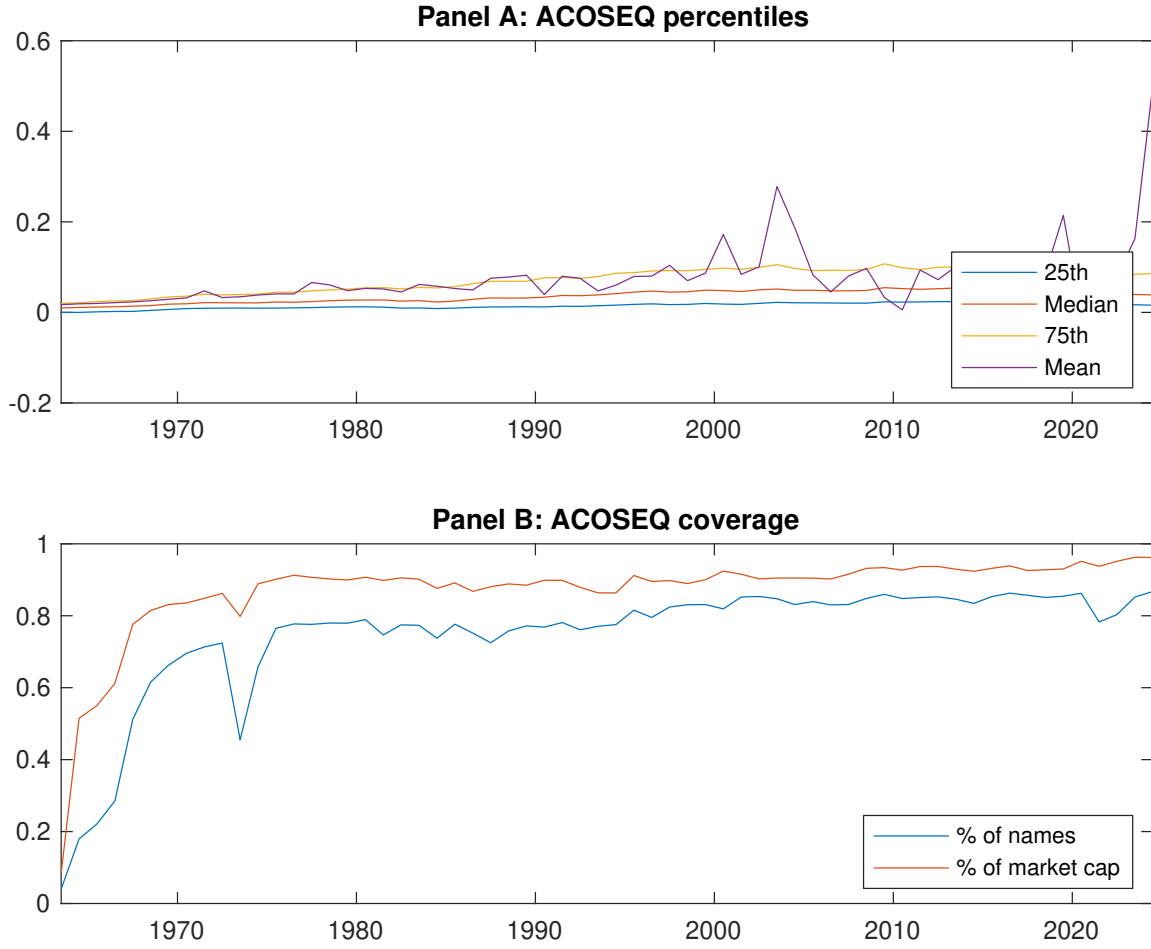


Figure 1: Times series of LLI percentiles and coverage. This figure plots descriptive statistics for LLI. Panel A shows cross-sectional percentiles of LLI over the sample. Panel B plots the monthly coverage of LLI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LLI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on LLI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.51 [2.80]	0.62 [3.58]	0.54 [3.27]	0.66 [3.96]	0.76 [4.34]	0.25 [3.01]
α_{CAPM}	-0.11 [-1.89]	0.03 [0.57]	-0.03 [-0.60]	0.09 [1.96]	0.16 [3.37]	0.27 [3.13]
α_{FF3}	-0.19 [-4.10]	0.06 [1.17]	0.01 [0.26]	0.13 [3.05]	0.18 [3.74]	0.37 [4.83]
α_{FF4}	-0.16 [-3.37]	0.04 [0.71]	0.04 [0.90]	0.11 [2.57]	0.16 [3.41]	0.32 [4.19]
α_{FF5}	-0.15 [-3.19]	0.10 [1.90]	0.01 [0.30]	0.06 [1.38]	0.08 [1.87]	0.24 [3.19]
α_{FF6}	-0.13 [-2.66]	0.07 [1.43]	0.04 [0.84]	0.05 [1.13]	0.08 [1.78]	0.21 [2.79]
Panel B: Fama and French (2018) 6-factor model loadings for LLI-sorted portfolios						
β_{MKT}	1.08 [92.79]	0.99 [81.43]	0.94 [87.61]	0.98 [96.00]	1.02 [93.27]	-0.06 [-3.26]
β_{SMB}	-0.12 [-7.01]	-0.00 [-0.20]	0.03 [1.64]	-0.00 [-0.02]	0.11 [6.98]	0.23 [8.74]
β_{HML}	0.28 [12.51]	-0.03 [-1.29]	-0.13 [-6.14]	-0.15 [-7.67]	-0.11 [-5.27]	-0.39 [-11.26]
β_{RMW}	-0.09 [-4.03]	-0.07 [-2.89]	0.01 [0.31]	0.16 [7.98]	0.22 [10.19]	0.31 [8.77]
β_{CMA}	-0.04 [-1.18]	-0.09 [-2.74]	0.00 [0.13]	0.09 [2.95]	0.10 [3.06]	0.13 [2.61]
β_{UMD}	-0.03 [-2.96]	0.03 [2.68]	-0.03 [-3.24]	0.01 [1.41]	0.00 [0.38]	0.04 [2.13]
Panel C: Average number of firms (n) and market capitalization (me)						
n	711	795	690	608	601	
me (\$10 ⁶)	1616	1882	1910	2219	2051	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LLI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.01]	0.27 [3.13]	0.37 [4.83]	0.32 [4.19]	0.24 [3.19]	0.21 [2.79]
Quintile	NYSE	EW	0.17 [2.59]	0.12 [1.97]	0.10 [1.66]	0.11 [1.72]	0.00 [0.06]	0.02 [0.27]
Quintile	Name	VW	0.26 [3.03]	0.28 [3.18]	0.39 [4.88]	0.34 [4.19]	0.24 [3.16]	0.21 [2.72]
Quintile	Cap	VW	0.20 [2.44]	0.23 [2.78]	0.32 [4.21]	0.30 [3.88]	0.18 [2.44]	0.17 [2.35]
Decile	NYSE	VW	0.18 [1.76]	0.20 [1.92]	0.30 [3.29]	0.32 [3.45]	0.22 [2.37]	0.25 [2.61]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [2.77]	0.25 [2.92]	0.34 [4.42]	0.32 [4.09]	0.22 [2.97]	0.20 [2.76]
Quintile	NYSE	EW	0.01 [0.08]					
Quintile	Name	VW	0.24 [2.81]	0.26 [2.99]	0.36 [4.50]	0.33 [4.13]	0.22 [2.98]	0.21 [2.74]
Quintile	Cap	VW	0.18 [2.19]	0.20 [2.50]	0.28 [3.76]	0.27 [3.60]	0.16 [2.20]	0.16 [2.15]
Decile	NYSE	VW	0.15 [1.51]	0.18 [1.70]	0.27 [2.91]	0.28 [3.03]	0.18 [2.02]	0.20 [2.16]

Table 3: Conditional sort on size and LLI

This table presents results for conditional double sorts on size and LLI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LLI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LLI and short stocks with low LLI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LLI Quintiles					LLI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.54 [2.20]	0.70 [2.67]	0.81 [3.22]	0.91 [3.44]	0.73 [2.73]	0.19 [2.13]	0.13 [1.48]	0.08 [0.94]	0.11 [1.22]	0.04 [0.40]	0.06 [0.69]
	(2)	0.65 [2.86]	0.75 [3.13]	0.82 [3.51]	0.73 [3.03]	0.89 [3.70]	0.24 [2.64]	0.19 [2.19]	0.17 [1.87]	0.21 [2.35]	0.08 [0.86]	0.12 [1.36]
	(3)	0.76 [3.75]	0.64 [3.04]	0.77 [3.54]	0.88 [3.92]	0.87 [3.88]	0.12 [1.26]	0.05 [0.57]	0.03 [0.30]	0.04 [0.38]	-0.06 [-0.64]	-0.05 [-0.48]
	(4)	0.60 [3.08]	0.57 [2.91]	0.75 [3.83]	0.86 [4.32]	0.78 [3.84]	0.18 [2.09]	0.15 [1.76]	0.21 [2.53]	0.20 [2.38]	0.11 [1.33]	0.11 [1.33]
	(5)	0.50 [2.71]	0.60 [3.64]	0.47 [2.89]	0.61 [3.78]	0.72 [4.27]	0.23 [2.35]	0.26 [2.71]	0.39 [4.48]	0.35 [3.90]	0.25 [2.93]	0.22 [2.58]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LLI Quintiles					LLI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	380	389	387	386	379	31	32	32	31	28	
	(2)	104	105	105	104	104	54	54	55	54	54	
	(3)	74	74	74	74	74	94	92	95	95	96	
	(4)	62	62	62	62	62	204	203	205	205	212	
(5)	56	56	56	56	56	1354	1543	1539	1700	1617		

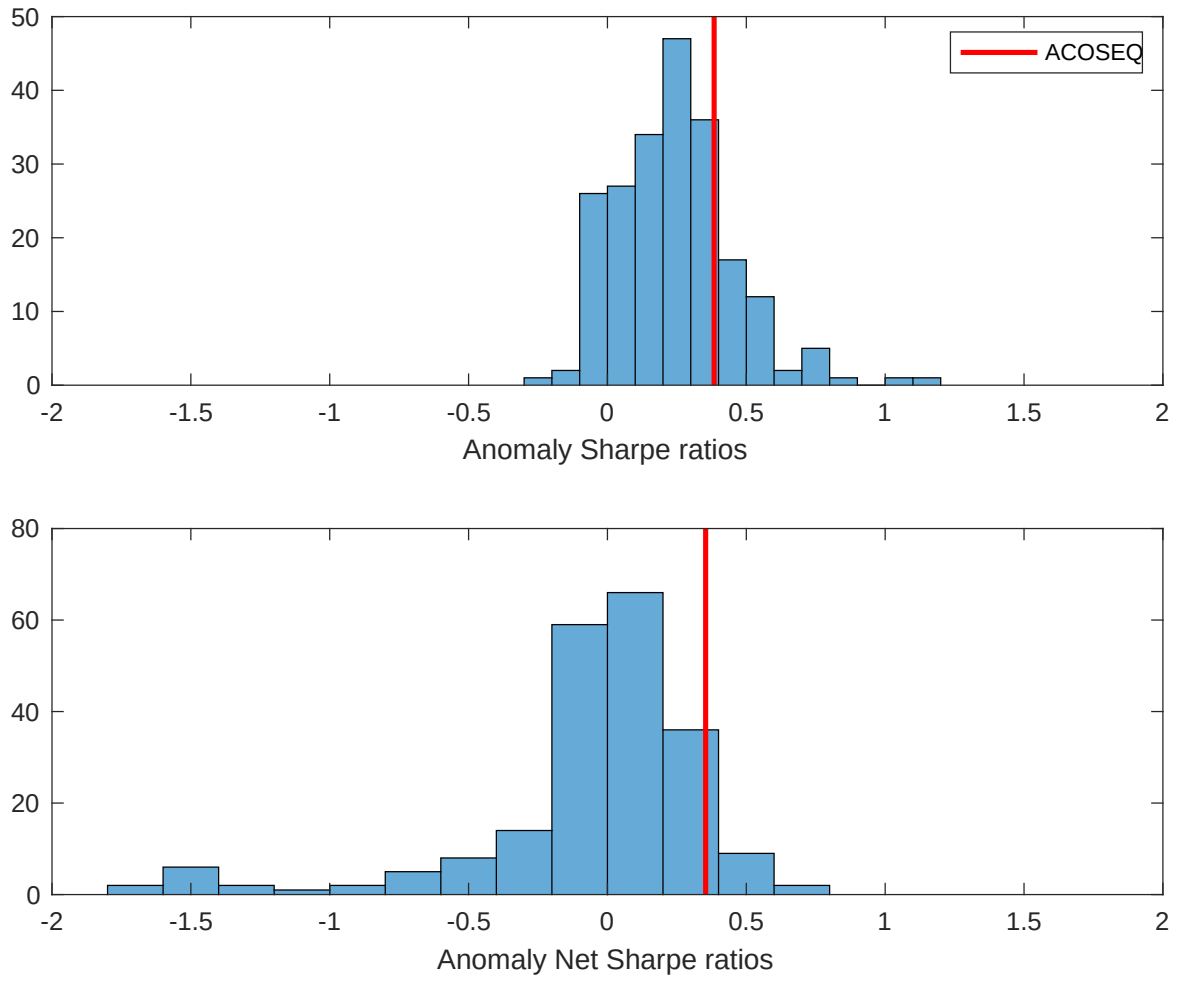


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LLI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

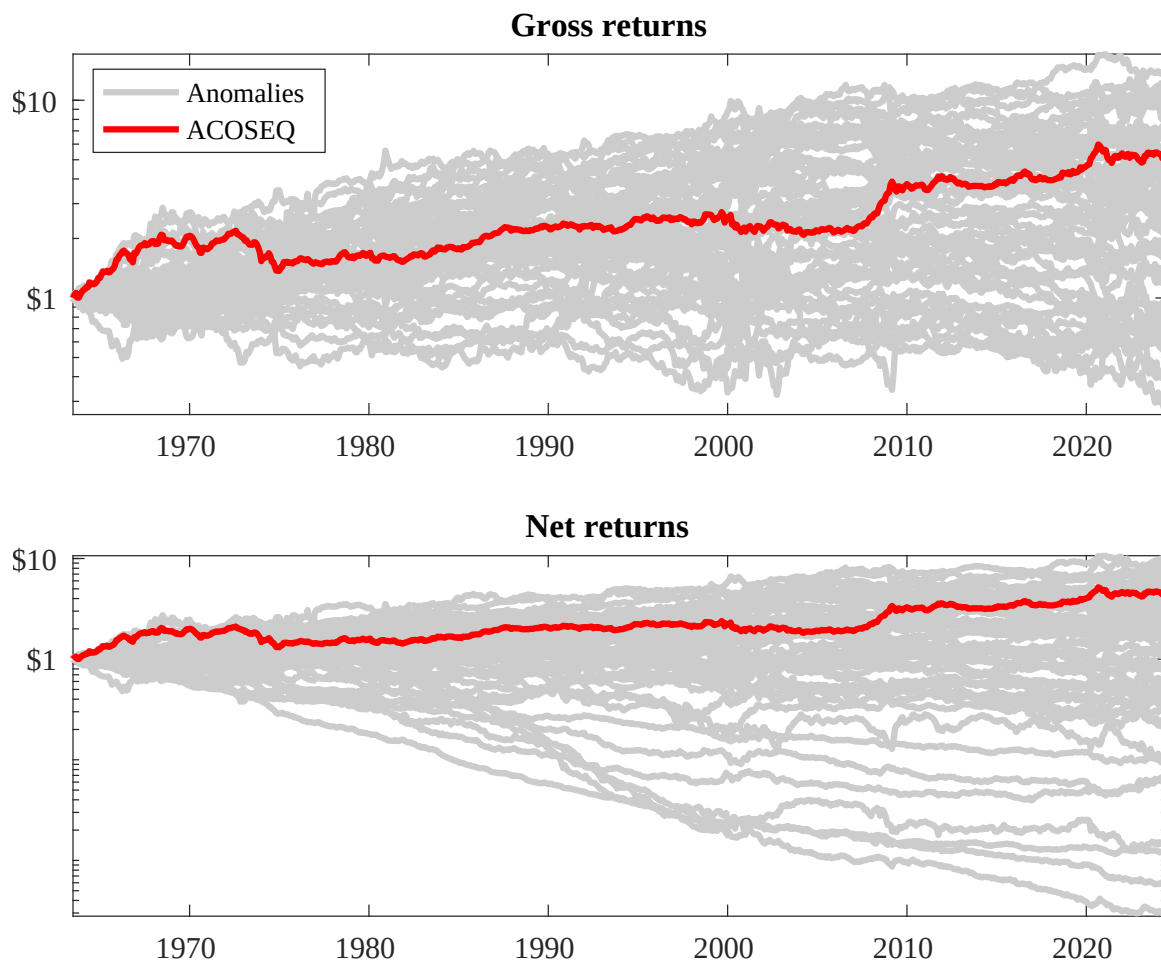
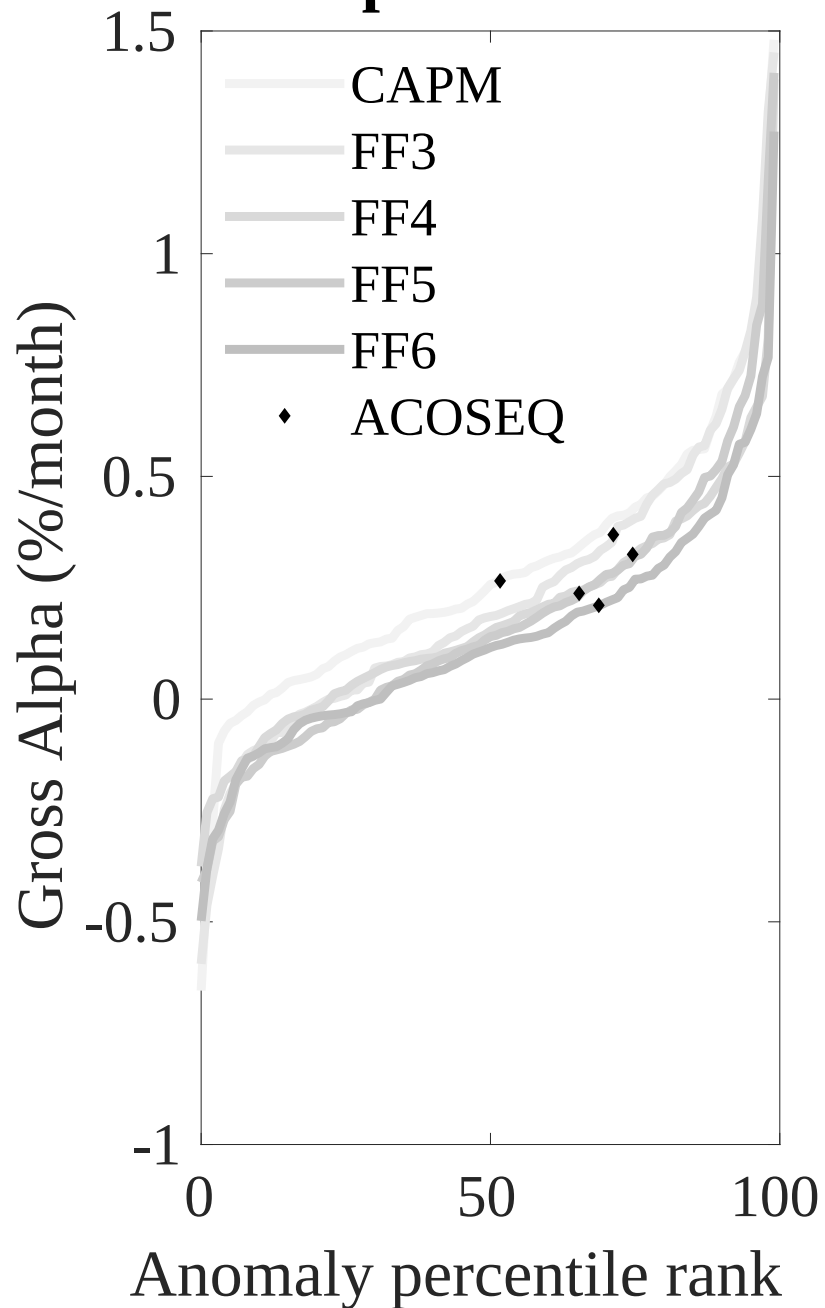


Figure 3: Dollar invested.

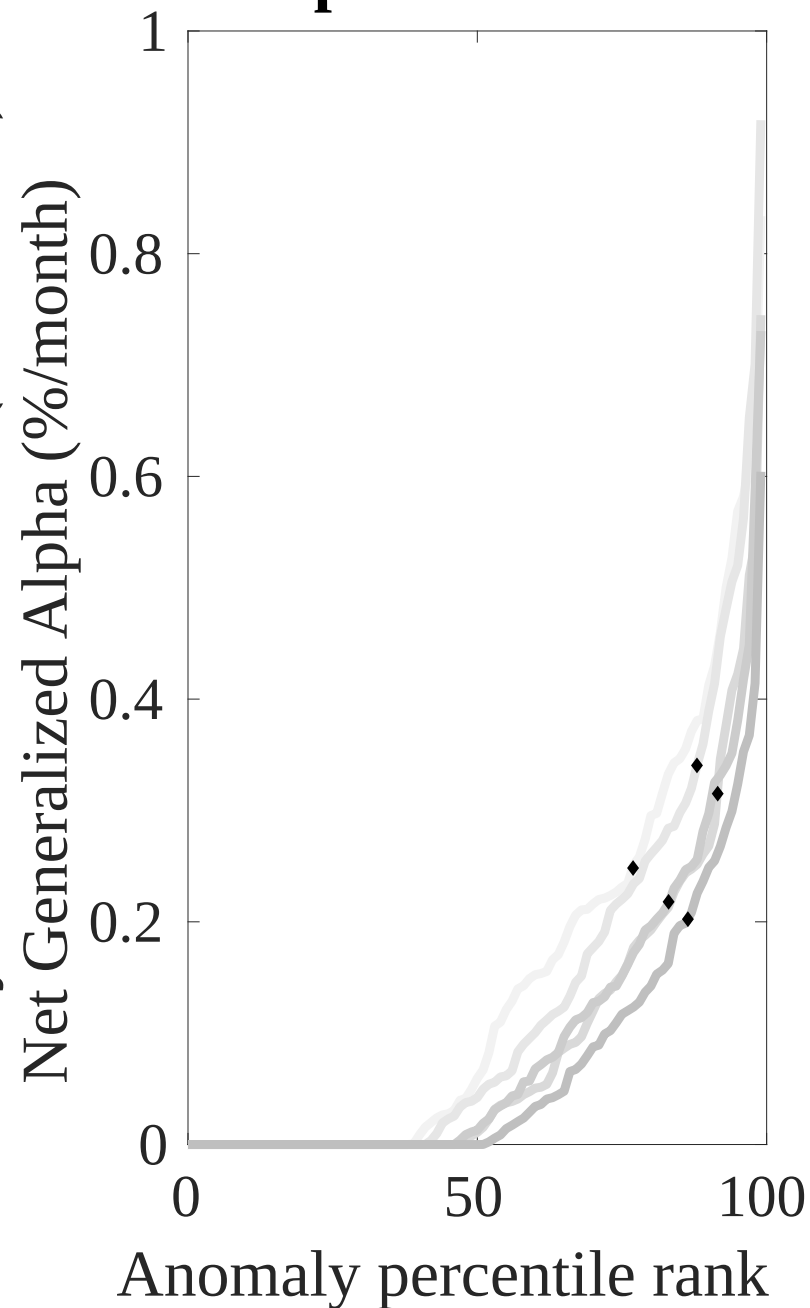
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LLI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

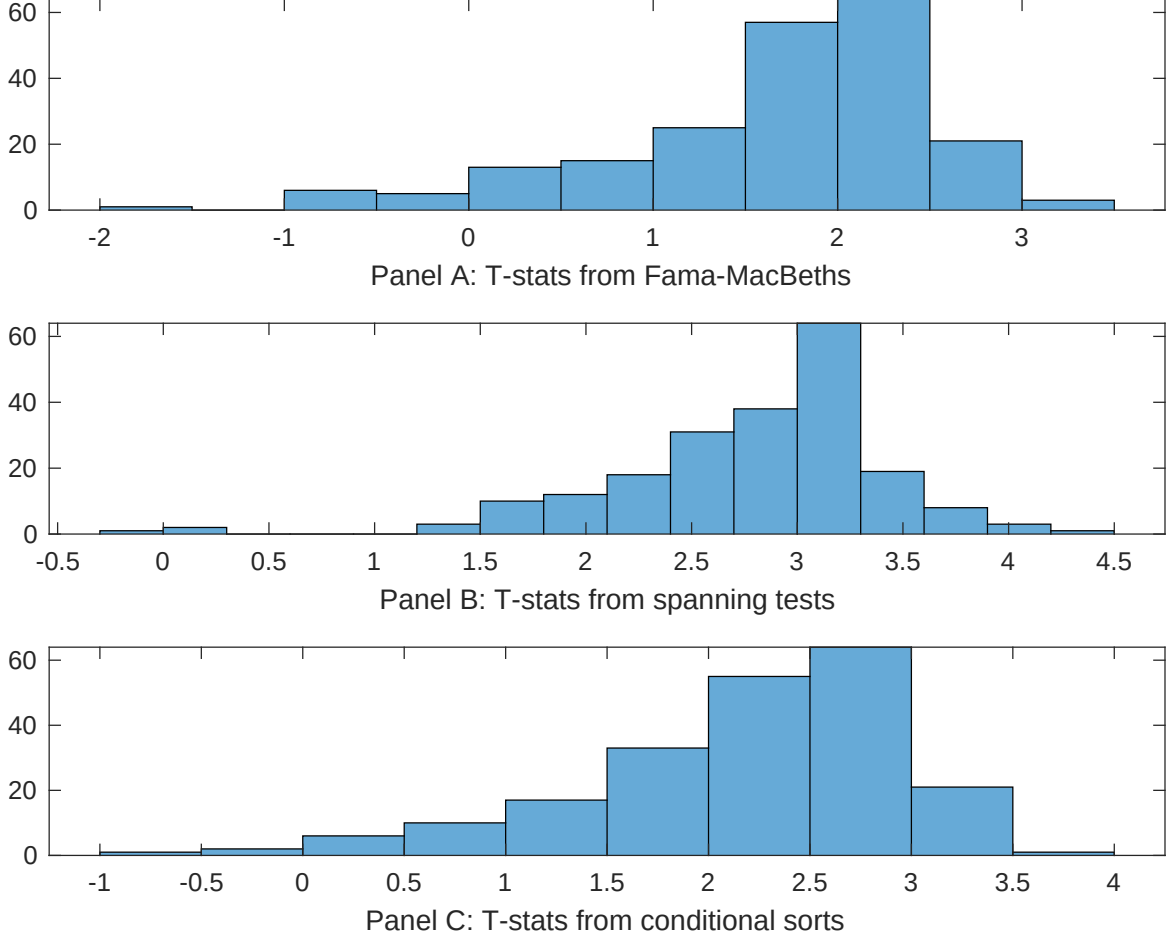


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LLI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LLI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LLI}LLI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LLI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LLI. Stocks are finally grouped into five LLI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LLI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LLI}LLI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Operating leverage, gross profits / total assets, Book to market, original (Stattman 1980), Book leverage (annual), Enterprise component of BM, Equity Duration. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.10 [4.40]	0.92 [3.70]	0.14 [6.25]	0.12 [5.09]	0.11 [4.81]	0.20 [9.08]	0.17 [7.61]
LLI	0.57 [1.60]	0.67 [1.75]	0.12 [2.94]	0.90 [2.75]	0.82 [2.24]	0.11 [3.10]	0.69 [2.33]
Anomaly 1	0.13 [3.58]						0.66 [0.02]
Anomaly 2		0.68 [4.90]					0.64 [4.77]
Anomaly 3			0.37 [7.37]				0.26 [4.66]
Anomaly 4				0.13 [1.31]			-0.43 [-1.83]
Anomaly 5					0.60 [4.79]		0.36 [2.32]
Anomaly 6						0.51 [5.26]	0.43 [3.93]
# months	720	720	715	720	715	720	715
$\bar{R}^2(\%)$	0	0	1	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LLI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LLI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Operating leverage, gross profits / total assets, Book to market, original (Stattman 1980), Book leverage (annual), Enterprise component of BM, Equity Duration. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17 [2.38]	0.14 [1.85]	0.19 [2.48]	0.18 [2.40]	0.25 [3.38]	0.20 [2.70]	0.17 [2.37]
Anomaly 1	27.99 [9.78]						22.12 [6.95]
Anomaly 2		21.69 [7.21]					14.05 [4.07]
Anomaly 3			-19.68 [-4.53]				-10.70 [-2.30]
Anomaly 4				10.37 [3.70]			0.96 [0.35]
Anomaly 5					-17.13 [-4.65]		-20.41 [-5.65]
Anomaly 6						-10.30 [-2.65]	4.57 [1.10]
mkt	-5.57 [-3.27]	-6.19 [-3.54]	-5.14 [-2.86]	-4.52 [-2.47]	-4.18 [-2.30]	-5.72 [-3.17]	-3.41 [-1.99]
smb	10.73 [3.86]	21.90 [8.63]	30.73 [10.09]	22.67 [8.72]	24.69 [9.50]	26.26 [9.25]	16.64 [5.28]
hml	-23.67 [-6.53]	-29.14 [-8.04]	-18.08 [-3.23]	-31.41 [-7.94]	-28.66 [-7.16]	-26.90 [-4.82]	-2.85 [-0.48]
rmw	15.34 [4.17]	18.23 [4.76]	27.74 [7.81]	33.88 [9.39]	24.99 [6.76]	30.48 [8.66]	2.28 [0.53]
cma	7.40 [1.52]	16.22 [3.25]	14.48 [2.84]	12.81 [2.51]	9.63 [1.88]	10.74 [2.07]	9.04 [1.84]
umd	3.41 [2.02]	3.36 [1.93]	4.91 [2.75]	3.51 [1.96]	0.18 [0.09]	3.85 [2.15]	-0.87 [-0.47]
# months	732	732	728	732	728	732	728
$\bar{R}^2(\%)$	36	33	30	29	30	29	41

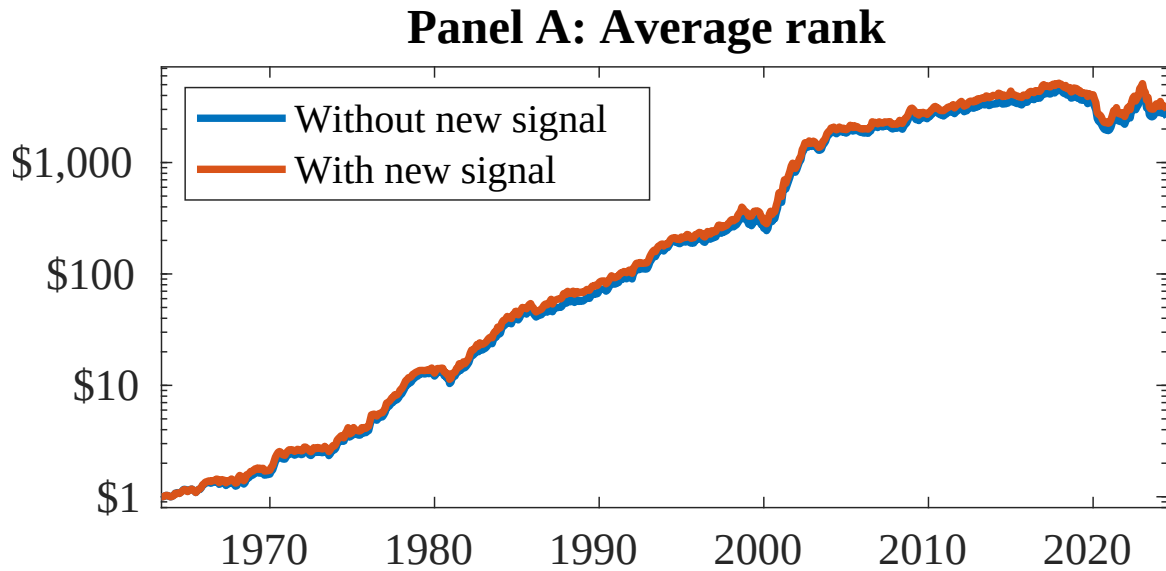


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as LLI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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